Elementary reductions for the fixed-target argument

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These are scoped analytic reductions, not a proof of the sharp iid Berry–Esseen conjecture. They were independently derived from the external suggestions supplied for the present revision. AI systems were used substantially in the derivation, checking and writing of this note. The current manuscript and its certificates are unchanged.

1 One fifth-moment inequality

Let X have mean zero and variance one, and assume $\mathbb{E}|X|^5 < \infty$. Write

$$m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad \rho = \mathbb{E}|X|^3, \quad M_j = \mathbb{E}|X|^j,$$
$$\beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad \delta = \gamma - \rho\beta.$$

Both strict signs have positive probability, so $|\beta| < 1$. An atom at zero is allowed. Put

$$P_s = \frac{(\rho+\gamma)(\delta)_+^2}{8(1+\beta)^2} + \frac{(\rho-\gamma)(-\delta)_+^2}{8(1-\beta)^2}.$$

Proposition 1. *Under these hypotheses,*

$$\frac{\rho M_4}{2} - \frac{M_5}{4} \le \frac{\rho^3}{4} - P_s - \frac{\delta^2}{16m}. \tag{1}$$

Its loss dominates each of the three former losses

$$P_s, \qquad \frac{\delta^4}{4\rho(1-\beta^2)^2}, \qquad \frac{\delta^2}{8m}.$$

Proof. For one sign let $Y = X_+$ or X_- and put $a = \mathbb{E}Y^2$, $b = \mathbb{E}Y^3$, $d = b - \rho a$, and $V = \mathbb{E}[Y^3(Y-\rho)^2]$. Cauchy gives

$$d^2 = \{\mathbb{E}[Y^{1/2}Y^{3/2}(Y-\rho)]\}^2 \le mV.$$

If $d \ge 0$, the fourth and fifth moments $c = \mathbb{E}Y^4$ and $e = \mathbb{E}Y^5$ satisfy $c \ge b^2/a$ and $e \ge c^2/b$. Hence

$$V = e - 2\rho c + \rho^2 b \ge (c-\rho b)^2/b \ge b(b/a-\rho)^2 = bd^2/a^2.$$

The second inequality uses $c \ge b^2/a \ge \rho b$; it would not follow by squaring a negative lower bound.

On the two signs,

$$a_\pm = (1 \pm \beta)/2, \quad b_\pm = (\rho \pm \gamma)/2, \quad d_\pm = \pm\delta/2.$$

Use the stronger bound on the sign with $d_\pm \ge 0$, and the bound $V_\mp \ge d_\mp^2/m$ on the other sign. Since

$$V_+ + V_- = M_5 - 2\rho M_4 + \rho^3,$$

division by four proves (1). If $\delta = 0$ both losses vanish and the same identity proves the claim.

For domination, suppose first that $\delta > 0$. Signwise Cauchy gives $mb_+ \geq a_+^2$, so $P_s = b_+\delta^2/(16a_+^2) \geq \delta^2/(16m)$. Thus the combined loss dominates $\delta^2/(8m)$. Also

$$\delta/2 = b_+a_- - b_-a_+ \leq b_+a_-, \qquad b_+ \leq \rho,$$

so $\delta^2 \leq 4\rho b_+a_-^2$. As $1 - \beta^2 = 4a_+a_-$, this proves $P_s \geq \delta^4/[4\rho(1-\beta^2)^2]$. The case $\delta < 0$ is the same calculation with the signs exchanged. □

The new addition charges different signs of one nonnegative square. It does not justify adding the old full weighted loss to P_s. For example, the variance-one centered law

$$\mathbb{P}(X = -1/4) = 396/425, \quad \mathbb{P}(X = 4) = 99/1700,$$
$$\mathbb{P}(X = -1/2) = 1/125, \quad \mathbb{P}(X = 2) = 1/500$$

violates the assertion with loss $P_s + \delta^2/(8m)$: its excess over the available square is greater than 0.0267. The new loss is smaller than that square by more than 0.0201.

2 The first moment from one determinant

Proposition 2. *Suppose additionally $|X| \leq R$. Necessarily $1 \leq \rho \leq R$ and*

$$2m \leq U(R,\rho) := \frac{R + \rho + \sqrt{(R-\rho)(4R^3 - 3R - \rho)}}{2R^2}. \tag{2}$$

This upper bound equals

$$\inf_{a>0} \frac{R - \rho + 2a + Ra^2}{a(2R + a)}. \tag{3}$$

It is nondecreasing in R and nonincreasing in ρ on $1 \leq \rho \leq R$. Therefore a closed parameter cell may use $U(R_{\text{upper}}, \rho_{\text{lower}})$, with its square root rounded upwards.

Proof. For $Y = |X|$ and $u = \mathbb{E}Y$, the matrix

$$\mathbb{E}\left[(R - Y)\begin{pmatrix}1\\ Y\end{pmatrix}\begin{pmatrix}1 & Y\end{pmatrix}\right] = \begin{pmatrix}R - u & Ru - 1\\ Ru - 1 & R - \rho\end{pmatrix}$$

is positive semidefinite. Its determinant gives

$$-R^2u^2 + (R + \rho)u + R^2 - R\rho - 1 \geq 0.$$

The discriminant is $(R-\rho)(4R^3 - 3R - \rho)$; solving the quadratic proves (2). The radicand is nonnegative, since $R \geq \rho \geq 1$ and $4R^3 - 3R - \rho \geq 4R(R^2 - 1) \geq 0$.

For completeness, the infimum identity can also be checked without a formal minimization. For $1 < R$ and $1 \leq \rho < R$, the upper root U satisfies $1/R < U \leq 1 < R$. Indeed the determinant polynomial is positive at $u = 1/R$, nonpositive at $u = 1$, and decreases for $u \geq 1$. At

$$a_* = (RU - 1)/(R - U) > 0$$

the quadratic

$$(R - U)a^2 + 2(1 - RU)a + (R - \rho)$$

has zero minimum, because its determinant vanishes. It is nonnegative for every a. Dividing by $a(2R + a) > 0$ proves that the expression in (3) is at least U, with equality at a_*. When

$\rho = R > 1$, its infimum as $a \downarrow 0$ is $1/R = U$. When $R = 1$, also $\rho = 1$ and the expression is identically one.

For fixed $a > 0$ the expression decreases in ρ. Its derivative in R has the sign of $a^3 - 3a + 2\rho \geq -2 + 2\rho \geq 0$. Taking infima proves both monotonicity statements, including endpoints. □

This replaces the eleven sampled auxiliary values of a by their exact optimal value. It does not replace the stronger signwise matrix constraints; those remain separate valid information about the same law.

3 A calculus proof for the cosine constant

Proposition 3. *For every* $0 \leq x \leq 1$,

$$1 - \cos(2\pi x) \leq \frac{4553}{1000} x.$$

Proof. For $y > 0$ put $h(y) = (1 - \cos y)/y$. The derivative has the sign of $N(y) = y \sin y + \cos y - 1$. On $(0, \pi)$ its sign is the sign of $y - \tan(y/2)$. The function $\tan(y/2) - y$ decreases up to $\pi/2$ and increases thereafter to infinity; it is negative at $\pi/2$. It therefore has exactly one zero y_* in $(\pi/2, \pi)$. On $(\pi, 2\pi)$, $N(y) < 0$. Thus the maximum of h on $[0, 2\pi]$ is attained at y_*, and the half-angle identity gives

$$h(y_*) = \frac{2y_*}{1 + y_*^2}.$$

Take $y_0 = 23311/10000$. The six-term alternating lower bound is

$$N(y_0) > \sum_{j=1}^{6} \frac{(-1)^{j-1}(2j-1)y_0^{2j}}{(2j)!} > \frac{3}{200000} > 0. \tag{4}$$

The alternating remainder has the asserted sign: the ratio of successive term magnitudes is $y_0^2/[(2j-1)(2j+2)]$, which is below one for every $j \geq 6$ and decreases. Since $\pi/2 < y_0 < \pi$, this proves $y_* > y_0$. The function $y/(1 + y^2)$ decreases for $y > 1$. Using $\pi < 355/113$, we get

$$\max_{0<x\leq 1} \frac{1 - \cos(2\pi x)}{x} = \frac{4\pi y_*}{1 + y_*^2} < \frac{4(355/113)y_0}{1 + y_0^2} < \frac{4553}{1000} - \frac{1}{10000}.$$

The last comparison and (4) are exact rational inequalities. To make the elementary constant enclosure reproducible, use $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and the alternating arctangent series, with eight and two terms respectively. Their outward intervals already prove $3 < \pi < 355/113$. Machin's identity follows by the tangent addition formula and the fact that the angle in question lies in $(0, \pi/2)$. The case $x = 0$ is immediate. □

The constant $23/5$ follows since $4553/1000 < 23/5$. Consequently both former Bernstein coefficient tables can be replaced by this one calculus argument and its few rational comparisons. This changes neither the harmonic span domain nor the other count-comparison hypotheses. The value $9/2$ is not a valid replacement constant: at $x = 3/8$ the ratio is $(8 + 4\sqrt{2})/3 > 9/2$.

4 A common two-sign transfer for every sample size

Here only $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\rho = \mathbb{E}|X|^3 < \infty$ are needed; bounded support is not required. Define probability measures on $(0, \infty)$ by

$$\mu_\pm(dy) = \frac{y}{m}\mathbb{P}(\pm X \in dy,\ \pm X > 0), \qquad c_\pm = \int y\, d\mu_\pm(y) > 0.$$

These are the distributions obtained by weighting each strict sign by its absolute magnitude. Define

$$Q = \frac{m}{c_-}\delta_{-c_-} + \left(1 - \frac{m}{c_-} - \frac{m}{c_+}\right)\delta_0 + \frac{m}{c_+}\delta_{c_+}, \qquad \eta = \rho(P) - \rho(Q).$$

Proposition 4. *Q is a probability measure with the same first two moments as P, and*

$$\eta = m\{\mathrm{Var}_{\mu_-} Y + \mathrm{Var}_{\mu_+} Y\} \geq 0. \tag{5}$$

For the characteristic functions ψ of P and g of Q,

$$|\psi(t) - g(t)| \leq \frac{|t|^3}{6}\eta \qquad (t \in \mathbb{R}). \tag{6}$$

For every integer $n \geq 1$,

$$|\psi(t)^n - g(t)^n - ng(t)^{n-1}(\psi(t) - g(t))| \leq \frac{n(n-1)}{2}|\psi(t) - g(t)|^2 \leq \frac{n(n-1)}{72}|t|^6\eta^2. \tag{7}$$

All statements include zero atoms and $\eta = 0$.

Proof. On each sign Cauchy gives $m^2 \leq \mathbb{P}(\pm X > 0)\mathbb{E}X_\pm^2 = \mathbb{P}(\pm X > 0)mc_\pm$. Thus $m/c_\pm \leq \mathbb{P}(\pm X > 0)$ and the mass at zero in Q is nonnegative. Its strict-sign first and second moments equal those of P: they are m and $mc_\pm$. Its third absolute moment is $m(c_-^2 + c_+^2)$, whereas that of P is $m\int y^2(d\mu_- + d\mu_+)$. This proves (5). If $\eta = 0$, both $\mu_\pm$ are point masses at $c_\pm$; the original law is then exactly Q, including its possible zero atom.

For $\sigma \in \{-1, 1\}$ put

$$h_{\sigma,t}(y) = \frac{e^{i\sigma ty} - 1}{y} = i\sigma t\int_0^1 e^{i\sigma tsy}\, ds,$$

with the continuous definition at zero. The integral representation gives

$$h''_{\sigma,t}(y) = (i\sigma t)^3\int_0^1 s^2 e^{i\sigma tsy}\, ds, \qquad |h''_{\sigma,t}(y)| \leq |t|^3/3.$$

Taylor's formula along the real segment from c_σ to y yields

$$h(y) - h(c_\sigma) - h'(c_\sigma)(y - c_\sigma) = (y - c_\sigma)^2\int_0^1 (1 - r)h''(c_\sigma + r(y - c_\sigma))\, dr.$$

Every integral is justified by the finite second moment of μ_σ. The linear term integrates to zero. Finally,

$$\psi(t) - g(t) = m\sum_{\sigma\in\{-1,1\}}\left\{\int h_{\sigma,t}(y)\, d\mu_\sigma(y) - h_{\sigma,t}(c_\sigma)\right\}.$$

Taking absolute values proves (6). The same two measures occur at every frequency; no independent frequencywise law has been introduced.

For $n \geq 2$, the segment joining the two characteristic functions lies in the closed unit disk. Taylor's formula for z^n along this segment bounds the second remainder by $n(n-1)|\psi - g|^2/2$. For $n = 1$ the remainder is zero. This proves (7). □

For the normalized sum, replace t by $t/\sqrt{n}$ in (7); the last bound becomes $(n-1)|t|^6\eta^2/(72n^2)$. This is an all-n transfer, not a uniform CDF estimate: inversion, smoothing reserves and any signed first-order improvement remain separate requirements.

The coefficient $1/6$ in (6) is sharp. For $0<\varepsilon<1/6$, take masses $1/2-\varepsilon, 3\varepsilon, 1/2-3\varepsilon, \varepsilon$ at $-1,0,1,2$. This law is centered with variance one. Only the positive size-biased sign has nonzero variance, so $\mathbb{E}_P X^3 - \mathbb{E}_Q X^3 = \eta > 0$. Its characteristic difference has leading term $-it^3\eta/6$, with absolute remainder at most $|t|^4(\mathbb{E}_P X^4 + \mathbb{E}_Q X^4)/24$. Thus the proposed smaller coefficient $1/7$ fails whenever $0<|t|<4\eta/[7(\mathbb{E}_P X^4 + \mathbb{E}_Q X^4)]$.

Remark 5 (What the transfer does not replace)**.** *The reference Q can have three atoms and is not automatically Bernoulli. The modulus bound alone cannot supply a favorable signed Fourier response, a sharp finite-base comparison, or the stronger zero-mass radius used in some existing numerical estimates. The earlier response and price identities must remain wherever those extra constraints are used. The present formulation replaces only the scalar defect/response/remainder construction and supplies a common version for every n.*

5 One compact domain for the finite base

This last reduction uses an external inequality, unlike the elementary moment and transfer propositions above. Set $c=(3+\sqrt{10})/(6\sqrt{2\pi})$ and let $\Delta_n(P)$ denote the Kolmogorov distance from the normalized iid sum to the standard normal.

Proposition 6. *Assume the universal affine bound*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right).$$

For $1\le n\le 41$, every attaining fixed-target pair (P,H) with

$$a = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho \ge 0$$

satisfies

$$\rho < \frac{7361}{4000} < 1.841, \qquad \operatorname{supp} P \subset (-13,13), \qquad |H|<26.$$

No predecessor inequality is assumed.

Proof. The moment bound follows from $c > 819/2000$ and direct division:

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The fixed-target contamination inequality is

$$\begin{aligned} I(y) = n^{3/2}(h(y)-F) + ny\phi(H/\sqrt{n}) + \tfrac{1}{2}H\phi(H/\sqrt{n})(y^2-1) \\ + \tfrac{3}{2}c\rho(y^2-1) - c(|y|^3 - \rho - 3y\beta) \le 0, \end{aligned}$$

with equality on the closed support, where $F=\mathbb{P}(S_n\le H)$ and $h(y)=\mathbb{P}(S_{n-1}\le H-y)$. For $n=1$ the same contamination calculation has $h(y)=\mathbf{1}_{\{y\le H\}}$; no deleted random summand is required. The usual fixed-n attainment argument also applies at $n=1$: its tightness, moving closed threshold and third-moment penalty steps do not require $n\ge 2$. Thus the finite-base extremal reduction has no exceptional first index.

Let $u=|y|\ge 1$. We have $|\beta|\le 1$, $\phi(0)<2/5$, $\sup_z |z\phi(z)| = \phi(1) < 1/4$, $\sqrt{n}<13/2$, and $819/2000 < c < 41/100$. In particular $n^{3/2} < 533/2$ and

$$I(y) \le P(u) := 266.5 + 1.944715u^2 + 17.63u - 0.4095u^3.$$

To check the coefficients, the linear coefficient is $41(2/5)+3(41/100) = 17.63$ and the quadratic coefficient is $13/16 + (3/2)(41/100)(1841/1000) = 1.944715$. The constant contribution of the two $c\rho$ terms is $-c\rho/2$, which can be dropped; there is no additional positive constant to retain. The term with $H\phi$ is bounded by $(13/16)(u^2 - 1)$ before its negative constant is dropped. All displayed decimals are exact rationals.

Exact evaluation gives

$$P(13) < -75, \quad P'(13) < -139, \quad P''(13) < -28, \quad P''' = -2457/1000 < 0.$$

It follows successively that P'', P', P are negative on $[13, \infty)$. Closed-support contact therefore excludes both exterior rays, including their boundary points.

If $H < 0$, Chebyshev gives $F \le n/H^2$; if $H > 0$, the positive discrepancy is at most $1 - \Phi(H/\sqrt{n}) \le n/H^2$. Thus $|H| \ge 26$ would imply

$$\sqrt{n}(F - \Phi(H/\sqrt{n})) < \frac{533/2}{26^2} = \frac{819}{2000} - \frac{397}{26000} < c\rho,$$

contradicting $a \ge 0$. The elementary constants used above follow from the same directed π bounds as in the cosine proof, directed squares around $\sqrt{10}$, and $e > 8/3$, which gives $\phi(1) < 1/4$ from $\pi > 3$. □

This is one compact finite-index problem, not a solution of that problem. The reference law in Proposition 4 can still have three atoms, and a complete estimate of its sums and the original-law response remains necessary.

6 Verification boundary

The proofs above establish their stated analytic implications. The accompanying exact script checks rational coefficient identities, scalar signs and a fixed adversarial collection of actual centered variance-one laws, including atoms at zero, reflected skew laws, arbitrarily small mixture weights and a distant atom. It also rejects the false stronger fifth-moment sum and the false cosine and characteristic constants described above. Such tests are checks on the derivations, not substitutes for their proofs.

These reductions need no new dense parameter cover. Applying their stronger bounds to the full entry or local comparison still requires the original domain and coverage conditions. No old proof object is deleted, no target is weakened, and no complete finite-base or global conclusion is asserted.